



大连理工大学
DALIAN UNIVERSITY OF TECHNOLOGY



VEQ: 快速高保真的参数化平衡求解器

VEQ: a fast parametric Grad-Shafranov solver for fixed-boundary tokamak equilibria with flexible source inputs

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MHD 平衡是聚变等离子体物理分析的基础 (输运、稳定性等)

Analytic or engineering closures are inexpensive but provide compressed geometry and **limited control of shaping** and residual diagnostics.

需要比简化平衡模型保留更多二维几何, 例如 METIS

Full equilibrium or reconstruction workflows provide richer solver-native outputs but may be **too costly or infrastructure-heavy** for repeated inner-loop calls.

比传统的完整求解/重建流程更轻量, 例如 EFIT/CHEASE...

A compact continuous equilibrium representation that retains **useful two-dimensional shaping** and residual diagnostics while remaining **inexpensive** enough to regenerate in repeated calls.

传统代码的局限性

面向 “单次求解”

依赖固定的离散表示

新场景需求

先进模块: 面向反复、高频调用的基础算子

机器学习: 大规模参数扫描、数据生成

集成模拟: 要求更快的平衡位形求解

VEQ 平衡位形

数学: 参数化、光滑、可微 便于求导、重采样及 ML 耦合

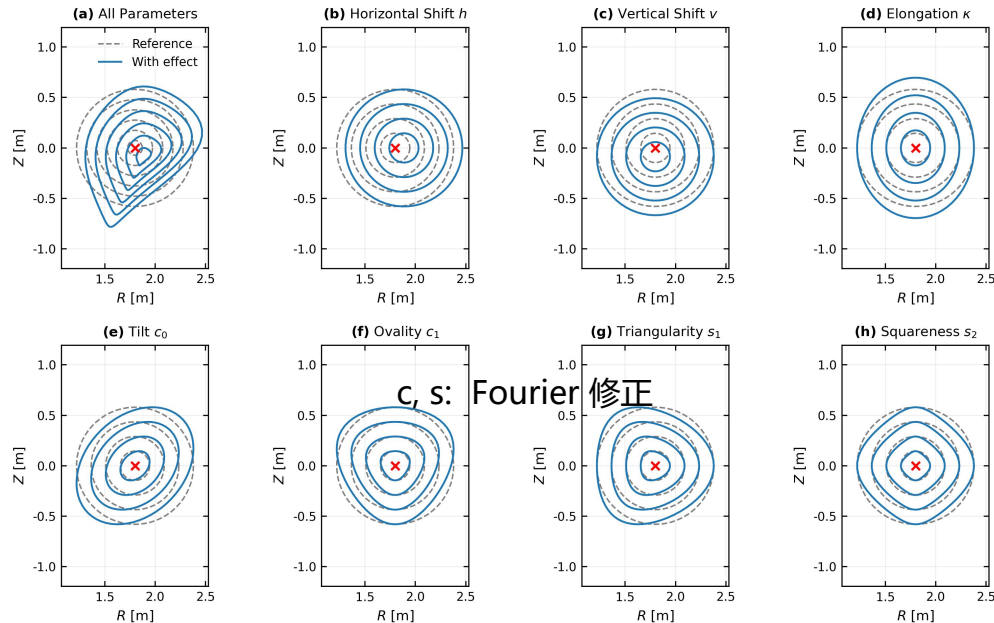
计算: 高保真 & 高效率并存 快速且大规模的数据生成

未知量是形状参数, 而非 $\psi(R, Z)$

$$\begin{aligned} R(\rho, \theta) &= R_0 + a [h(\rho) + \rho \cos \theta], \\ Z(\rho, \theta) &= Z_0 + a [v(\rho) - \rho \kappa(\rho) \sin \theta]. \end{aligned}$$

h, v : 磁面的位移

κ : 拉长比



c, s : Fourier 修正

极向: Fourier / MXH shaping

$$\begin{aligned} \bar{\theta}(\rho, \theta) &= \theta + c_0(\rho) + \sum_{m=1}^M c_m(\rho) \cos(m\theta) \\ &\quad + \sum_{n=1}^N s_n(\rho) \sin(n\theta), \end{aligned}$$

径向: shifted-Chebyshev profiles

$$\begin{aligned} x &= 2\rho^2 - 1, \\ T_0(x) &= 1, \quad T_1(x) = x, \\ T_{l+1}(x) &= 2xT_l(x) - T_{l-1}(x). \end{aligned}$$

$$\begin{aligned} h &= (1 - \rho^2) \sum_{l=0}^L h_l T_l(x), \\ v &= (1 - \rho^2) \sum_{l=0}^L v_l T_l(x), \\ \kappa &= \kappa_a + (1 - \rho^2) \sum_{l=0}^L \kappa_l T_l(x), \\ c_m &= \rho^m \left[c_{ma} + (1 - \rho^2) \sum_{l=0}^L c_{ml} T_l(x) \right], \\ s_n &= \rho^n \left[s_{na} + (1 - \rho^2) \sum_{l=0}^L s_{nl} T_l(x) \right], \end{aligned}$$

变分原理:

$$\mathcal{L} = \int L dV, \quad dV = 2\pi JR d\rho d\theta, \quad L = \frac{|\nabla\psi|^2}{2\mu_0 R^2} - \left(\frac{F^2}{2\mu_0 R^2} + P \right).$$

得到 Grad-Shafranov 算子:

$$\mathcal{G} = -\mu_0 \left[JR \frac{\partial L}{\partial \psi} - \frac{\partial}{\partial \rho} \left(JR \frac{\partial L}{\partial \psi_\rho} \right) - \frac{\partial}{\partial \theta} \left(JR \frac{\partial L}{\partial \psi_\theta} \right) \right].$$

$$= \frac{J}{R} (FF_\psi + \mu_0 R^2 P_\psi) + \left(\frac{g_{\theta\theta} \psi_\rho}{JR} \right)_\rho - \left(\frac{g_{\rho\theta} \psi_\rho}{JR} \right)_\theta.$$

根据物理关系自然闭
合到 GS 中

1. 六种物理输入

2. 支持 rho/psi 的剖面 and 导数输入

Route	Direct input	Recovered for residual
PF	P_ψ, FF_ψ	ψ_ρ
PP	P_ψ, ψ_ρ	FF_ψ
PI	P_ψ, I_{tor}	ψ_ρ, FF_ψ
PJ1	P_ψ, j_{tor}	$I_{\text{tor}}, \psi_\rho, FF_\psi$
PJ2	$P_\psi, j_{\parallel}$	$I_{\text{tor}}, \psi_\rho, FF_\psi$
PQ	P_ψ, q	ψ_ρ, FF_ψ

对流残差为0:

$$\delta_{\mathbf{r}} \psi + \nabla \psi \cdot \delta \mathbf{r} = 0, \quad \delta \mathbf{r} = \sum_k \frac{\partial \mathbf{r}}{\partial p_k} \delta p_k.$$

变分弱形式:

$$\int_0^1 \int_0^{2\pi} \mathcal{G} \left(\frac{\partial \psi}{\partial R} \frac{\partial R}{\partial p_k} + \frac{\partial \psi}{\partial Z} \frac{\partial Z}{\partial p_k} \right) d\theta d\rho = 0.$$

主方法: $N_x \rightarrow N_x$

标准 GS 残差 (后处理/诊断):

$$\mathcal{G}_{\text{std}} = \frac{R}{J} \mathcal{G} = \Delta^* \psi + FF_\psi + \mu_0 R^2 P_\psi,$$

强形式伪谱配点法 (DESC 等):

$N_r * N_t \rightarrow N_x$

$$\min_{\mathbf{x}} \frac{1}{2} \|\mathcal{W}^{1/2} \mathbf{g}_{\text{std}}(\mathbf{x})\|_2^2.$$



VEQPy

Fixed-boundary Grad-Shafranov solver

VEQPy:

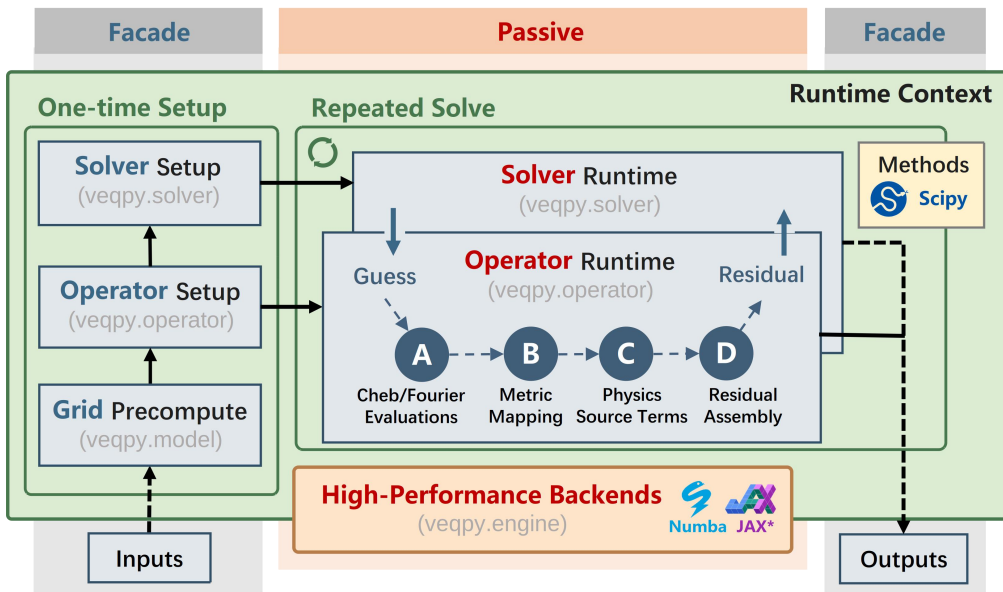
A Python package for tokamak MHD equilibrium calculations, serving the **VEQ (Veloce EQuilibrium)** project.

code:

github.com/zhangtakeda/veqpy

arXiv:

doi.org/10.48550/arXiv.2606.11821



Backend: Numba 默认, C++ 可选

Façade

(表层的易用性--包装、输入输出处理):

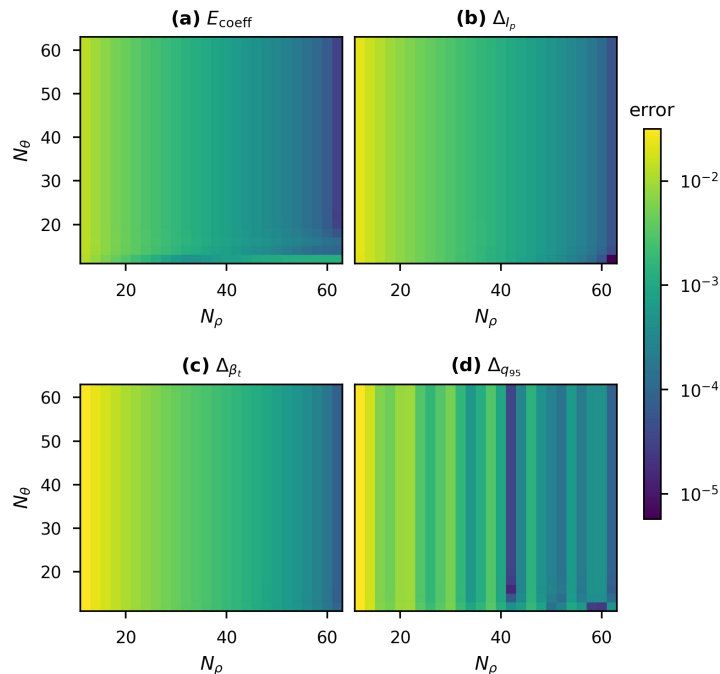
1. 输入检查 and 无量纲化
2. 内存分配 and 预计算

Passive

(尽可能避免开销--预计算, 预分配, 重复调用):

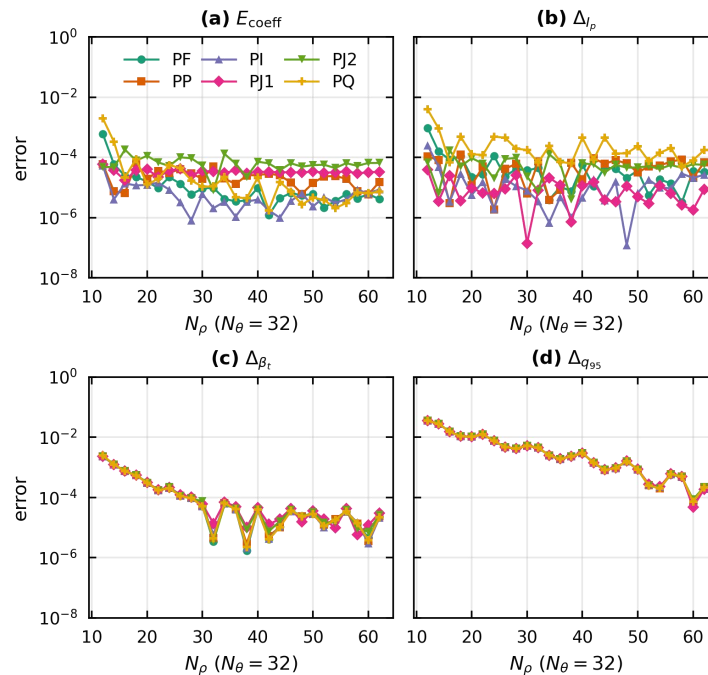
1. 在已有内存块上执行计算
2. 在 [配置, 模式] 组合中执行具体逻辑

径向和极向分辨率上的网格收敛行为



数值稳定性: 随着 N_ρ 、 N_θ 增大, 误差稳定下降, 说明后处理与诊断是可靠的。

六种输入路径之间的一致性

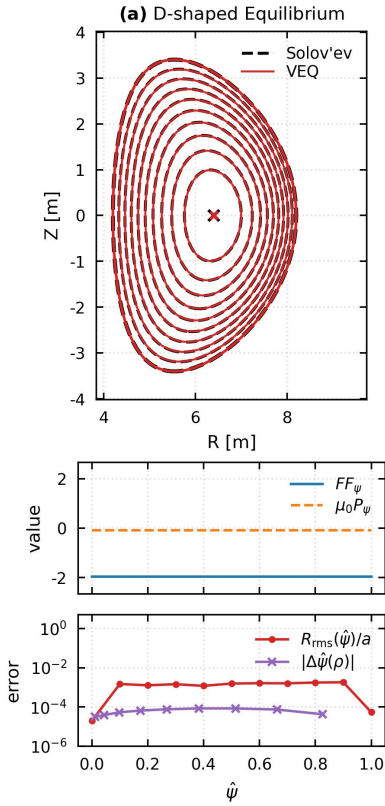


物理一致性: 六种输入路径共享同一有限维残差系统, 且在核心物理量上保持良好一致性。

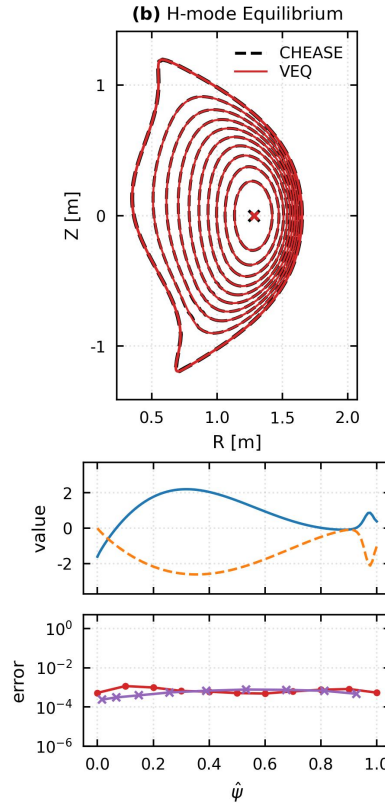
实际应用：多种等离子体位形求解



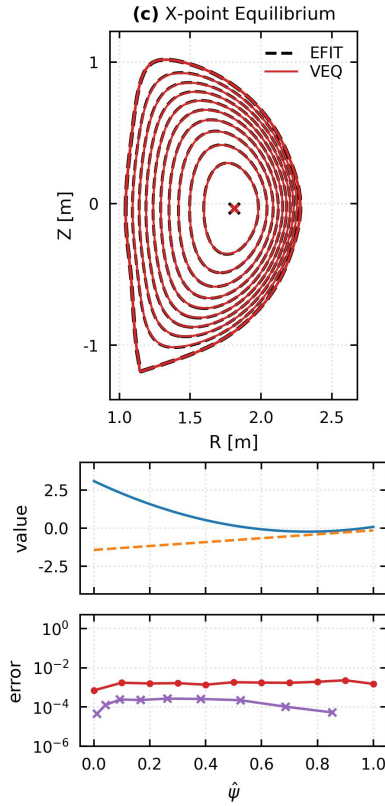
D-shape (75 参数)
Solovev 解析解



H-mode (130 参数)
CHEASE(有限元)



X-point (130 参数)
EFIT(差分/Green)



Case	Params	Time [ms]
D-shape	75	6.95
H-mode	130	115.62
X-point	130	23.40

Case	E_{gqdisk}/a	E_{lcfs}/a
D-shape	1.41×10^{-3}	1.95×10^{-4}
H-mode	7.33×10^{-4}	1.06×10^{-3}
X-point	1.69×10^{-3}	4.55×10^{-4}

D-shape: 75 参数, 对比解析解
H-mode: 130 参数, 对比 CHEASE
X-point: 130 参数, 对比 EFIT

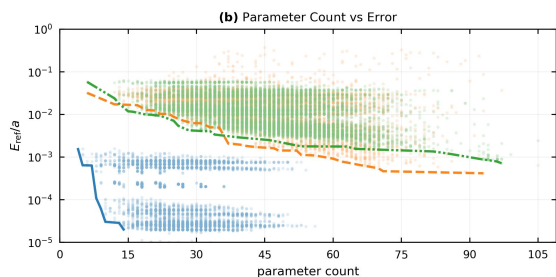
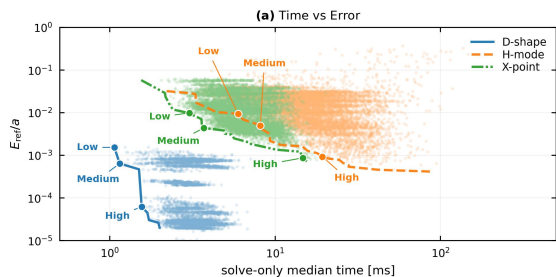
复杂位形测试提供了可信基础

实际应用：参数量对误差的影响

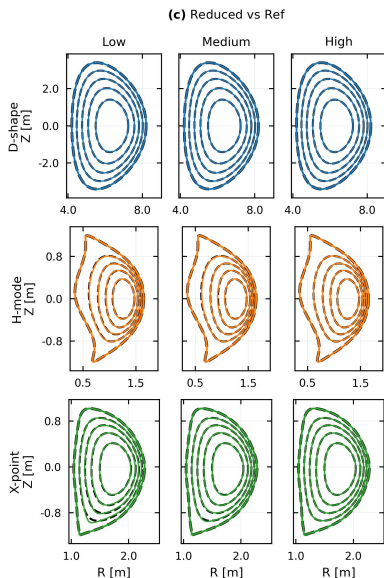


参数数量增加时，几何误差整体下降，但求解成本上升

reduced-VEQ 配置的
精度-成本 Pareto 分析



和 Ref-VEQ 的精度差异



VEQ 可以根据任务需求主动选择合适的参数量：
任务驱动选择 Low / Medium / High / Ref

14600KF, Ubuntu22.04, ...

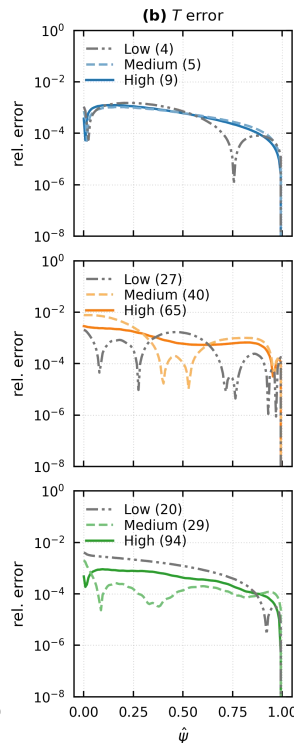
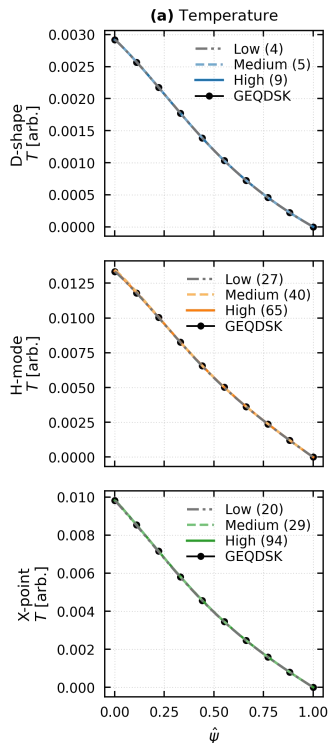
case(params)	Cxx (ms)	Numba (ms)	speedup
D-shaped(4)	0.172940	1.911798	11.055x
D-shaped(5)	0.214899	2.136862	9.944x
D-shaped(9)	0.229824	2.547320	11.084x
D-shaped(75)	1.000413	7.288244	7.285x
H-mode(27)	0.707322	5.654250	7.994x
H-mode(36)	0.844900	7.095212	8.398x
H-mode(60)	1.705001	15.468430	9.072x
H-mode(130)	8.265414	43.981732	5.321x
X-point(19)	0.374766	3.517772	9.387x
X-point(29)	0.549881	4.886525	8.887x
X-point(94)	1.975305	11.051075	5.595x
X-point(130)	4.267138	24.079790	5.643x

成本增加，但仍适合 repeated solve

1-D steady heat transport, 只考察几何通道误差传播

温度

相对误差



$$-\frac{d}{d\hat{\psi}} \left[V' \langle |\nabla \hat{\psi}|^2 \rangle \frac{dT}{d\hat{\psi}} \right] = V' S(\hat{\psi}),$$

$$T(1) = 0, \quad \left. \frac{dT}{d\hat{\psi}} \right|_{\hat{\psi}=0} = 0,$$

$$S(\hat{\psi}) = C \exp \left[- \left(\frac{\hat{\psi} - 0.18}{0.33} \right)^2 \right] (1 - 0.15\hat{\psi}),$$

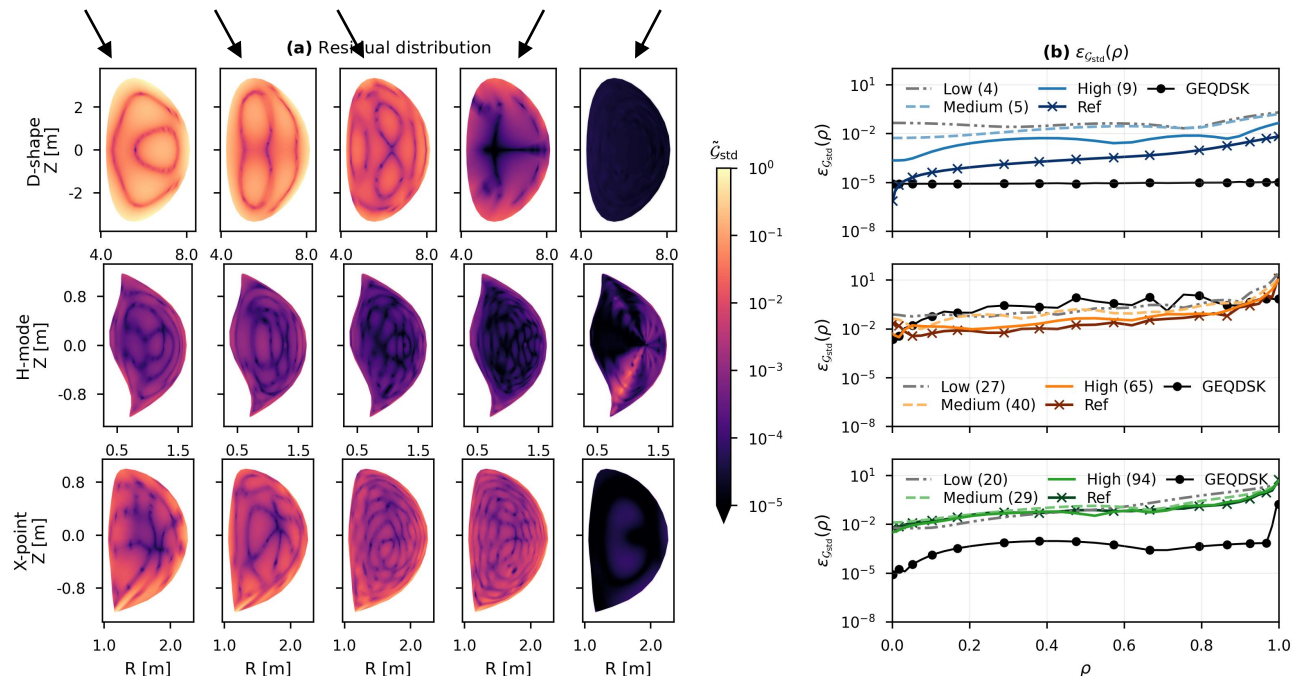
与 GEQDSK 耦合的结果非常接近

VEQ 的几何误差对输运模块完全可以接受

适合快速高保真的集成模拟

Case	RMS(δ_T)
D-shaped (4)	8.69×10^{-4}
D-shaped (5)	6.54×10^{-4}
D-shaped (9)	7.36×10^{-4}
H-mode (27)	8.57×10^{-4}
H-mode (40)	3.14×10^{-3}
H-mode (65)	1.34×10^{-3}
X-point (20)	1.68×10^{-3}
X-point (29)	3.18×10^{-4}
X-point (94)	5.36×10^{-4}

(Low, Medium, High)reduced,Ref-VEQ, GEQDSK 的采样局部强形式残差诊断

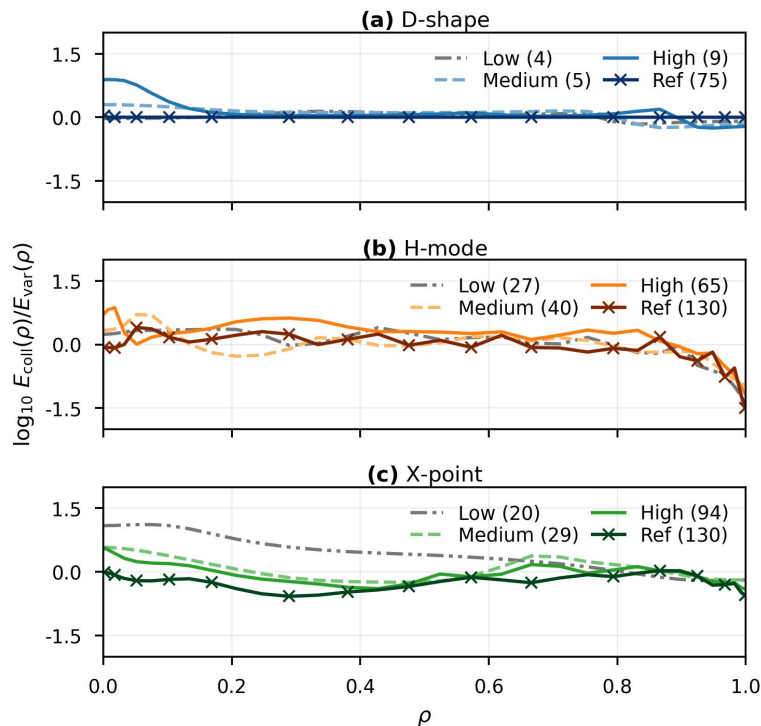


高维参数化确实提升了对参考平衡的描述能力,
但边界附近的残差仍是主要限制

将边界参数纳入优化很可能改进这一点 (自由边界)

类似 DESC 的求解方法, 同一有限维表示, 不同优化目标: **变分法--形状, 配点法--局部残差**

配点法局部 GS 残差 / 变分法 GS 残差



配点法后处理会降低最大残差, 但会牺牲形状

Case (Params)	$r_{\text{RMS,all}}$	$r_{\text{RMS}, < 0.8}$	$r_{\text{RMS}, \geq 0.8}$	$r_{ G_{\text{std}} _{\text{max}}}$	$r_{E_{\text{gdsk}}}$
D-shape (4)	0.843	1.026	0.799	0.744	1.265
D-shape (5)	0.723	1.146	0.668	0.541	1.227
D-shape (9)	0.650	1.207	0.597	0.389	1.043
D-shape (75)	0.999	1.001	0.999	1.007	1.000
H-mode (27)	0.071	1.415	0.067	0.026	3.532
H-mode (40)	0.139	1.227	0.135	0.074	3.710
H-mode (65)	0.109	1.978	0.105	0.037	8.549
H-mode (130)	0.050	0.932	0.050	0.025	3.341
X-point (20)	0.668	1.045	0.635	0.647	4.022
X-point (29)	0.677	1.247	0.655	0.634	5.461
X-point (94)	0.433	1.062	0.428	0.444	6.438
X-point (130)	0.319	0.812	0.316	0.374	4.425

全局/最大 RMS 降低, 但形状误差明显放大

有限参数维数下, “残差最小” 并不总等于 “几何最好”

Highlights:

- ◆ A projected residual solve enables repeated fixed-boundary geometry **solves in milliseconds**.
- ◆ **Six source-input routes** share one finite-dimensional Grad-Shafranov residual system.
- ◆ **D-shape, H-mode and X-point benchmarks** show normalized geometry accuracy.
- ◆ **A simplified transport test** shows geometry errors propagate mildly in one dimension.

美国: GA 开发的 FUSE.jl 主动联系并吸收了 VEQ 作为一种求解平衡的方法 (v1.1.6 版本开始)

欧洲: 将 VEQ 耦合进入 ASTRA 的工作正在推进

Future Extensions:

- ◆ Extend VEQ implementations to **free-boundary problems**.
- ◆ Extend VEQ to **three-dimensional equilibria**.
- ◆ Couple VEQ to **1.5-D integrated modeling**, transport or source models.

轻量、快速高保真: 我们也在开发国产自主的集成模拟平台, 初步的成果以及 benchmark 非常理想



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瞬原科技
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VEQPy

Fixed-boundary Grad-Shafranov solver

GitHub: github.com/zhangtakeda/veqpy

arXiv: doi.org/10.48550/arXiv.2606.11821